Department of Geography

Exercise Earth, Soil, Surface I

Administration: PhD candidate Andrew Mitchell

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Researching a Landslide in the Heinersreuther Forst using Geomorphic Mapping, ERT and root analysis (Report)

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Abbreviations

FoS	Factor of Safety
ERT	Electrical Resistivity Tomography
DWD	Deutscher Wetterdienst
DTM	Digital Terrain Model
DEM	Digital Elevation Model
RMSE	Root Mean Square Error
RAR	Root Area Ratio
T_r	Tensile Strength
t_r	Total Tensile Strength

1. Introduction

Since April 2020 roadworks began on the Autobahn A70 in Thurnau, near Bayreuth in Upper Franconia. The pathway had to be changed because of a nearby landslide (Siebert and Wilfing 2018). In total the costs of the project amounted to 65,000,000 € (Autobahn GmbH 2024). Features of landslides can regularly be observed in the Franconian Alb because of favorable geologic conditions (Dräbing et al. 2023; Siebert and Wilfing 2018).

There are two types of landslides: Rotational and translational slides (Bierman and Montgomery 2014; Dikau et al. 2019). The speed of landslides varies from soil creep similar stages to sudden events. Because of their range they have the ability to shape entire landscapes (ibid.). Landslides consist, from top to bottom, of a head scarp, secondary scarps, depressions, fissures, and the front or toe which marks the end of a landslide (ibid.). An activation happens when the factor of safety (FoS) of the slope drops below 1. The FoS is influenced by geology, roots, and water content of the soil (Bierman and Montgomery 2014; Bischetti et al. 2009; Dikau et al. 2019; Dräbing et al. 2023).

Our research tries to find the extent of the landslide in the Heinersreuther Forst. Therefore, we applied geomorphic mapping to discover features of the landslide. The second research aim is the exploration of landslide activity. Electrical resistivity tomography (ERT) was applied to find hints of the activity. Following the introduction, the research area and the applied methods are introduced as the obligatory part of this report, before the produced results are presented and discussed. The report ends with a conclusion and an outlook for future investigations.

2. Research Area

The research area is located in Upper Franconia, northern Bavaria, Germany. It is approximately 10 km west of Bayreuth's city center (Figure 1). Geologically, the study area is situated in the Triassic and Jurassic outskirts of the Franconian Alb (Dräbing et al. 2023; Figure 1). On top there are the rocks of the Triassic in form of the obtusus clay formation, which consists of pyrite-enriched claystones with banked lime in between (Peterek and Schröder 2010). Interlocking with the obtusus clay formation is the gryphäen sandstone formation. The gryphäen sandstone formation contains primarily course calcareous sandstone and marl (ibid.). Below the lower Triassic stratigraphically ends with the Rhätolias, a sedimentary rock formation consisting of sandstones (Dräbing et al. 2023; Peterek and Schröder 2010; Siebert and Wilfing 2018). The Rhätolias forms a visible scarp in the field above the transition from the Triassic to the Jurassic (Figure 5a). Jurassic rocks are claystones with variable fractions of sandstone and marl, the corresponding geologic formation is called Feuerletten (ibid.). Because of the geologic properties of claystones, the Feuerletten functions as an aquifer. Therefore, they function as sliding planes for landslides in the Rhätolias (Dräbing et al. 2023; Siebert and Wilfing 2018).

Beside the scarp in the Rhätolias, there are other observable features of a landslide in the research area (Bierman and Montgomery 2014; Dikau et al. 2019). Along the head scarp (Figure 5a), there are multiple breaking-ins. Near the head scarp, depressions and fissures can be found (Figure 5b; Figure 5c). Towards the river in the valley floor multiple secondary scarps were recognized (Figure 5b). At the end of the landslide deposit area the front or toe of the landslide is partly visible (Figure 5d). Fallen and bent trees were scattered sporadically in the research area (Figure 5e). In general, the research area was wetter in the Feuerletten. Not only streams, but wetlands can be found below the Rhätolias-Feuerletten boundary, which is an indication of the aquifer function of the Feuerletten (Dräbing et al. 2023; Siebert and Wilfing 2018).

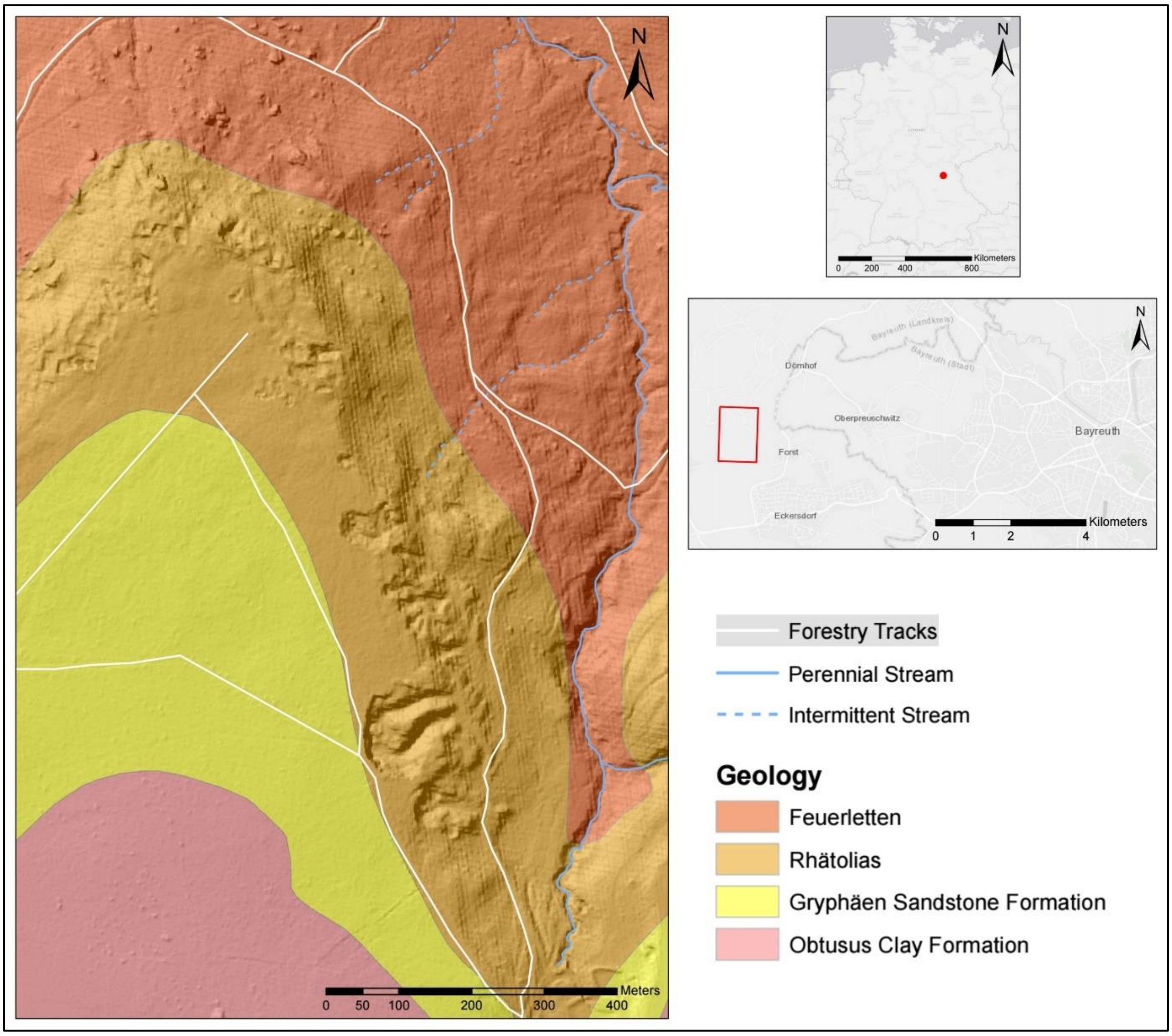

Figure 1: Location of the research area on the map of Germany, in relation to the city of Bayreuth, and the geology of the research area (Own Figure; Source: ESRI world contour map, geologic map of Bavaria (Online: UmweltAtlas (bayern.de))

February of 2024 had an increased precipitation in Bavaria in comparison to previous years (DWD 2024). Figure 2 shows the amount of daily precipitation in mm for the climate station in Oberwaiz, the closest settlement to the research area. The highest precipitation was registered on the 7th and 8th of February respectively with 24 mm and 25 mm. Comparable sums of rain happened after our field days. While the highest rainfall volume before the excursion on the 20th and the 21st reached 9 mm on the 19th of February, the wet research area is best explained by the constant precipitation throughout February in combination with the related geology.

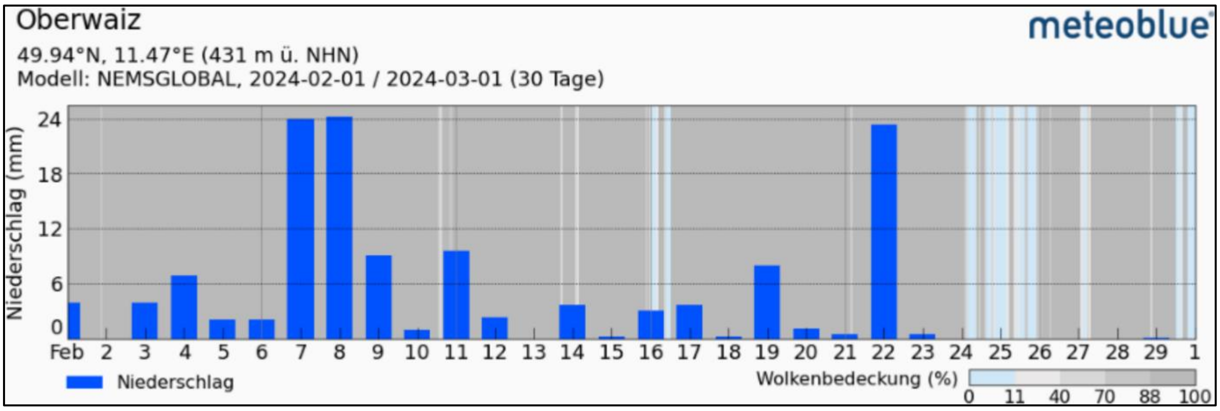

Figure 2: Daily precipitation in mm at the climate station in Oberwaiz in February of 2024 (Source: meteoblue, Wetterarchiv Oberwaiz - meteoblue, last visited 28.03.2024)

3. Methods

For this study we executed three methods: Geomorphic mapping, ERT, and root analysis. ERT can be split into more distinct parts of analysis of the topography, the measurement itself, the transect geology, and the FoS analysis. The applied methods will be discussed in detail in the following sections.

3.1 Geomorphic Mapping

The scope of a geomorphic map depends on the researched area as well as the study's aim (Seijmonsbergen 2013). Since the aim of the study is to assess the landslide area and its features, the geomorphic map is focused on them. On a short hike accompanied by PhD candidate Andrew Mitchell landslide features were mapped in the field (head scarp, secondary scarps, breaking-ins, depressions, fissures, fronts, wetlands, streams). These were initially marked on a copy of the digital terrain model (DTM) and had to be translated into a map in the lab. Some features were added as well. By coloring the digital elevation model (DEM) multiple depressions were recognized. Slumped terrain parts nicely showed the occurrence of wetlands and matched the observations of the Heinersreuther Forst. Forestry tracks and streams were mapped additionally for easier situation. The presentation geared to the guidelines from Laser and Stäblein (1980) and recent landslide studies (e.g., Dräbing et al. 2023). Finally, the locations of the ERT profile with transect geology points, root pits, and pictures were added to the geomorphic map.

3.2 ERT

The ERT started by assessing the topography of the ERT profile, continued with its actual measurement, and ended with the documentation of the transect's geology as well as topography ERT based FoS analysis. These distinct parts of the ERT will be discussed in the upcoming sections.

3.2.1 Assessment of the Topography

The topography was recorded with an angle measurement tool. To ensure the best results it was always handled by the same person. We separated the topography into sections of similar gradient and used trees as reference points to measure the gradient. Observed angles and the length of sections were noted. The estimated length of each section with its gradient translates to the topography of our ERT profile. We noticed that our topography overestimates the actual topography. By comparing the results to values of the DEM a coefficient was developed, which then resulted in an accurate topography model. This was assured by comparing it to the DEM and results from Dräbing et al. (2023). The topography was then inserted into the DAT-file of the ERT profile. Therefore, it was possible to plot the topography of the ERT transect.

3.2.2 Measurement and Evaluation of the ERT

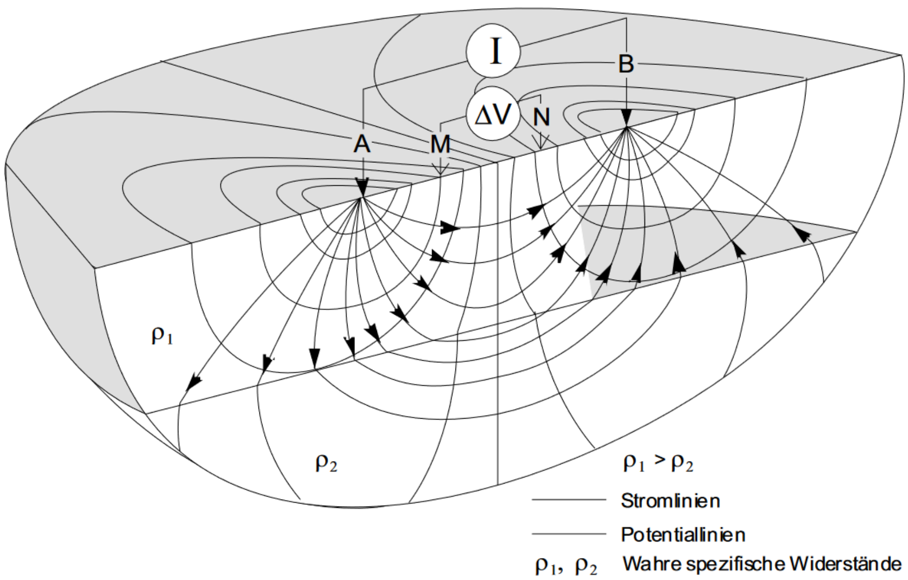

Figure 3: Principle of ERT. Electricity is inducted on the outer electrodes (A, B) and the resulting electric field is measured by the inner electrodes (M, N). This displays pseudo sections of the ground (Lange and Jacobs 2005: 129, Figure 5.20)

The ERT enables a cross-sectional view of the ground. It is often applied for the investigation of landslides because of its capability to differentiate between lithologies (Dräbing et al. 2023). Resistivities in the ground can be measured by applying a four-point measurement technique, which is presented in Figure 3. On the outer electrodes A and B electricity of a specific magnitude is inducted, which generates an electric field, its potential can then be detected by the inner electrodes M and N and translated to resistivities (Lange and Jacobs 2005). It is important to note that measured values are not an exact representation of lithology and are called pseudo sections. This results from inhomogeneities in the ground, which are averaged by the inversion (ibid.). Depending on geology it is possible to view up to 20 % of the length of the ERT profile into the ground (Gesellschaft für Geowissenschaftliche Untersuchungen 2011; Lange and Jacobs 2005).

In the Heinersreuther Forst we laid out four 100 m cables as straight as possible with a measuring tape. Beginning 25 m behind the head scarp, down the valley flank, through the valley floor, to the other side of the valley (Figure 4). The electrodes were connected with a 5 m spacing to the 100 m cables by jumper cables. The length of the profile should have reached 400 m, but the terrain shortened the transect to 385 m. The measurement started by placing the ABEM Terrameter LS 2 in between cable 1 and 2. The first measurement uses the command three-cables-first. For the next measurement the ABEM Terrameter LS 2 was placed in between cable 2 and 3, the corresponding command is one-cable-forward. For the last measurement, the ABEM Terrameter LS 2 was placed in between cable 3 and 4 with the command two-cables-last (ABEM Instrument AB 2016). There are parameters that need to be set before measuring (ibid.):

- Maximal voltage: 600 volts
- Maximal current: 200 milliampere
- Repetition of measurement: 2-3 times
- Electrode spacing: 5 m

Finally, an array needs to be set, which determines the mode of the measurement. The array for our measurements was the Wenner array. Wenner-array means that a constant electrode spacing is applied, while the Schlumberger array has changing electrode spacing for individual measurements (ABEM Instrument AB 2016).

To interpret the data, the results of the field measurements were inverted using the inversion software Res2DInv. Before the inversion, the assessed topography of the profile was inserted into the DAT-file produced by the ABEM Terrameter LS 2. The mesh setting in Res2DInv was changed to finest mesh. For the expected substantial difference in resistivities between the sandstones of the Rhätolias (high resistivity) and the clay stones of the Feuerletten (low resistivity) it produces the best results (Loke 2022). To further improve the accuracy of our results the node setting was changed to 4 nodes instead of 2 (ibid.). The robust inversion was selected to show sharp borders between lithological layers (ibid.). By default, 5 iterations of the inversion are calculated in Res2DInv, we added another 5 iterations, until the improvement in root mean square error (RMSE) dropped below 1 %. The least square method was used for the inversion (ibid.).

In the display section in Res2DInv the inversion results can be presented. As a scale we chose the automatic logarithmic scale of the program. Blueish values resemble values of up to 100 Ωm and indicate clays while yellow, orange, red, and purple values are an indication of sandstone with Ωm values above 900 Ωm (Dräbing et al. 2023).

3.2.3 Geology of the Transect

The geology of the transect was studied by taking out probes with a spate every 50 m along the ERT profile (Figure 4). The observable layering of the soil was described and will be presented in a later chapter with pictures (Figure 11). While a concrete description would be too long, differences are pointed out along the transect. Especially changes in lithology are presented and discussed with a focus on discrepancies between the Rhätolias and the Feuerletten.

3.2.4 FoS Analysis

The FoS analysis was used to determine the activity of the landslide area. It began by displaying the topography of the ERT and distinguishing between rotational and translational slide as well as separate slices. There are diverse ways of determining the FoS, we geared to the remarks of Selby (1993) and Bierman and Montgomery (2014).

Rotational Slide

The applied method for the rotational slide is the method of slices, which is based upon this formula:

$$\mathrm{FoS} = \frac{\sum_{i=1}^{n}[c'_{Si}\, l_i + (W_i \cos\beta_i - \mu_i\, l_i) \tan\emptyset_i]}{\sum_{i=1}^{n}[W_i \sin\beta_i]}$$

i is the number of slices, c'_S is the soil cohesion taken from Siebert and Wilfing (2018) with 48 kPa for the Rhätolias and 47.5 kPa for the Feuerletten, the base length of each slice l_i is calculated by the formula:

$$l_i = \frac{B_i}{\cos\beta_i}$$

B_i is the width of each slice, and β_i is the failure plane angle. W_i is the weight of each slice calculated with the formula:

$$W_i = \gamma_S\, z_i\, B_i$$

γ_S is the specific weight of soil, which is for both geologic units 21 kN/m³ (Siebert and Wilfing 2018), z_i is the depth of the soil for each slice. Moving back to the general formula of rotational slides, the next parameter is μ_i. μ_i is the pore water pressure and is described by the formula:

$$\mu_i = \gamma_W\, m\, z_i \cos(\beta)^2$$

γ_W is the specific weight of water with 9.81 kN/m³, and m is a coefficient resulting from the division of the height of the water table H by the slices total depth z_i. Because we did not have the value of the water table H, m was assumed to cover the values 0.3, 0.6, 0.9, and 1. Finally, $\emptyset_i$ is the angle of internal fraction taken from Siebert and Wilfing (2018) with 13.8 ° for the Rhätolias and 8.4 ° for the Feuerletten. We were able to determine the FoS per slice by inserting the values into the first formula. The sum of single rotational slides lead to a FoS for the entire slide section. A FoS below 1 describes an unstable stage while a FoS above 1 is a stable stage (Bierman and Montgomery 2014; Dikau et al 2019; Selby 1993).

Translational Slide

For the translational slide, the infinite slope method was applied, explained by this formula:

$$\mathrm{FoS} = \frac{c + [\rho_S - (\rho_W m)]\, g\, z_S \cos\beta \tan\emptyset}{\rho_S\, g\, z_S \sin\beta}$$

c is the cohesion, ρ_S is the density of the soil with 2141 kg/m³ by dividing the specific weight of soil by the gravitational acceleration g of 9.81 kg/m² (Siebert and Wilfing 2018), ρ_W is the density of water with 977 kg/m³, z_S is the thickness of the sliding slab, β is here the slope angle (Bierman and Montgomery 2014). Left over parameters were already explained in the section about rotational slides. As before were the distinct FoS for slices summed up to result in a FoS for the whole landslide type (Bierman and Montgomery 2014; Dikau et al 2019).

3.3 Root Analysis

For the root analysis we investigated three trees of the most present species in the Heiners-reuther Forst. Therefore, we found three representative adult trees for the species norway spruce, scots pine, and european beech. Then, a cube of the dimension 50*50*50 cm was excavated 1 m away from the stem downslope. The observable soil profile was documented and the locations of the so-called root pits are marked on the geomorphic map in Figure 4.

Root Area Ratio (RAR)

The RAR considers the side nearest to the tree. It was further separated into 5 sections of equal size (500 cm²). Roots with a diameter above 1 mm and below 10 mm were categorized to the closest mm and the number of roots per diameter was noted in the field (Bischetti et al. 2009). In the lab the RAR was evaluated by applying the formula:

$$\mathrm{RAR} = \frac{\sum_{i=1}^{i} A_R}{A}$$

A is the area investigated of 2500 cm², A_R is the area the roots between 1 and 10 mm take up in the soil. It can be calculated for each size class with the formula:

$$A_R = \left(\frac{\pi}{4}\right) d^2$$

The parameter d is the root diameter of each class. These were calculated in a matrix in excel, before summing them for each depth (0-10 cm, 10-20 cm, …). Results of the RARs for each section were then plotted using excel (Bischetti et al.2009).

Root Tensile Strength (T_r)

The excavated material was searched for roots of the target species. These were put into a bucket of water to ensure full saturation. The testing of their strength was performed by putting them in between two clamps . The measurement was performed by a scale mounted to the upper clamp. It was filmed during the test to ensure a per frame exact measurement at the moment of rupture. An optimal procedure was a rupture in or close to the middle of both clamps, if this was not the case, the test was ruled out.

The scale displayed the inflicted force in kg. Therefore, it needed to be converted to N by multiplying it with gravitational acceleration. The result of the calculation is the force at failure, which was then divided by the A_r to get the T_r in MPa. The results were then plotted and the R^2 was produced in excel.

Root Cohesion

$$t_r = \sum_{i=1}^{n} (T_r \, a_r)$$

This formula results in the total root tensile strength (t_r). a_r in this case is root area for each class individually. The result was then multiplied by two coefficients which correct for root deformation angles. According to Bischetti et al. 2009 the first coefficient is 1 and the second is 0.5. The result of the multiplication is the root cohesion, which was then plotted in excel.

4. Results

The results of our research will be presented in the same order as the methods. It will start with the geomorphic mapping, continue with the ERT and end with the root analysis before the results are discussed in chapter 5.

4.1 Results of the Geomorphic Mapping

The head scarp is recognizable on the DTM, DEM and in the field (Figure 5a). It is a significant step in the terrain. Secondary scarps are most obvious behind the head scarp as breaking-ins, which result of geomorphic processes after the initial landslide (Bierman and Montgomery 2014). They are also apparent below the head scarp as steps in the terrain (Figure 5b). There are four breaking-ins interrupting the head scarp. The furthest north located breaking-in is accompanied by fissures. These trenches derive from collapse settlements of the soil (ibid.). On a bigger scale collapse settlement can cause the formation of depressions (Bierman and Montgomery 2014). They are located below the head scarp or in between secondary scarps. At the end of the landslide towards the main stream the landslide front or toe was often incomplete (Figure 5d). None of the described features happened recently. However, the Rhätolias seems to inherit more geomorphic activity due to the occurrence of landslide features. They accumulate above the Rhätolias Feuerletten boundary, while below just some secondary scarps can be found and of course the landslide front. Simultaneously, wetlands and streams are seemingly restricted to the Feuerletten and do not appear in the Rhätolias.

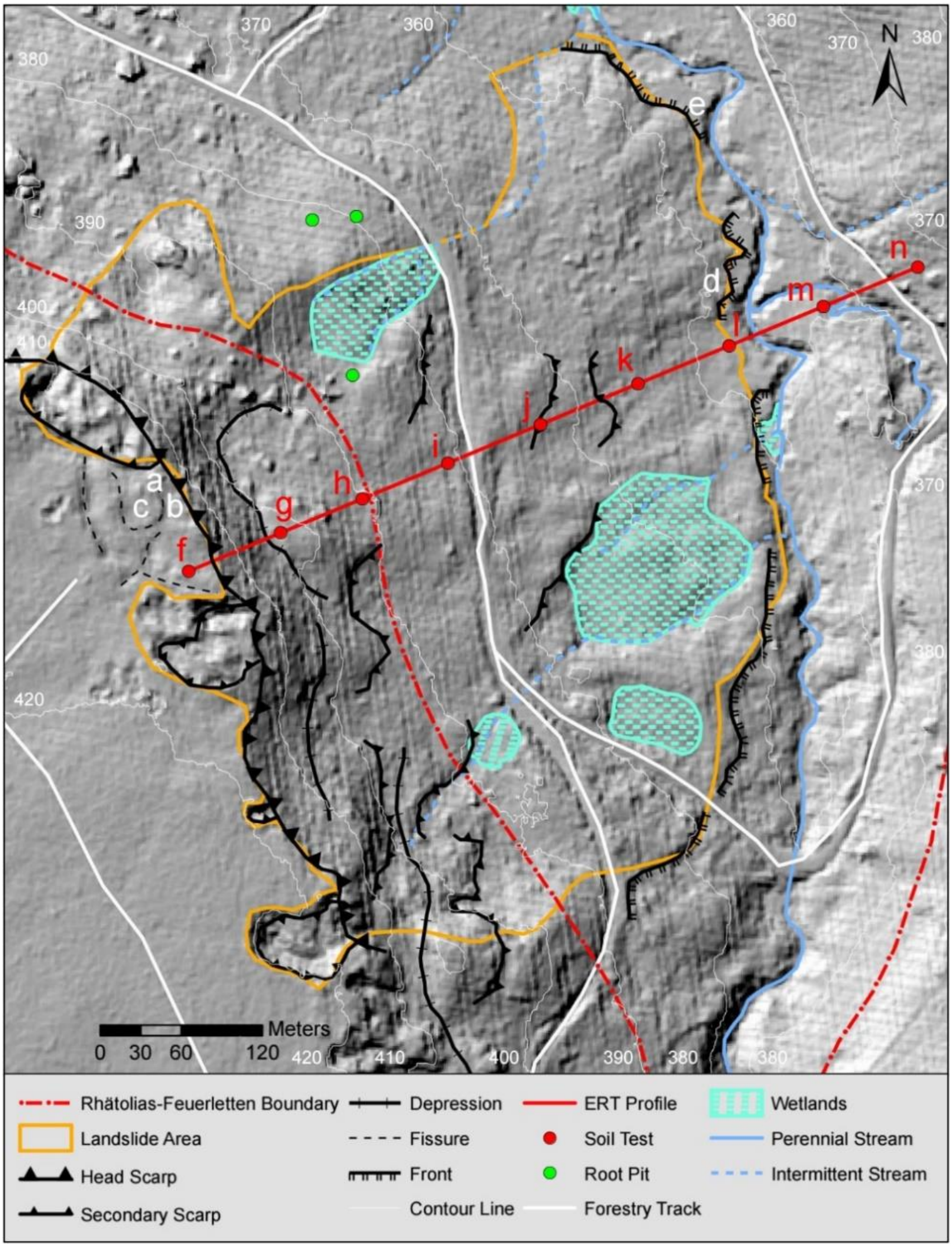

Figure 4: Geomorphic map of the landslide in the Heinersreuther Forst near Oberwaiz. Additional to features of the landslide, the Feuerletten Rhätolias boundary, roads, streams, and wetlands, the location of other methods is marked. The ERT profile, soil tests along the profile and the root pits are the situated methods. Letters in the geomorphic map correspond to pictures in figures below (Own Figure; Source: Geologic map of Bavaria (Online: UmweltAtlas (bayern.de))

Figure 5 shows landslide features in the field. In the first picture from the head scarp can be seen. Boulders lay on the surface with a downwards tilted layering (see Figure 5a). Figure 5b was taken on top of the head scarp towards a secondary scarp and a depression in between.

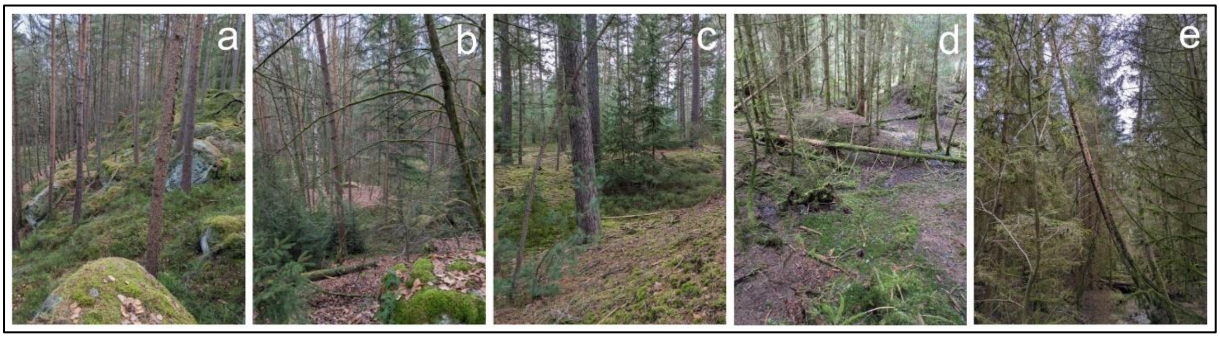

Figure 5: The head scarp (a), secondary scarp with depression (b), fissures above the head scarp (c), landslide front/toe of the landslide (d) in the Heinersreuther Forst. Additionally bent/fallen over trees near the river (e). The locations of the pictures are marked on the geomorphic map in Figure 4 (Own Figure; Source: Kevin Peter)

The already mentioned fissures behind the head scarp are visible in Figure 5c. Close to the stream in the valley floor the front of the landslide is partly recognizable (Figure 5d). Bent or fallen trees can be found more often near the stream (Figure 5e).

4.2 Results of the ERT

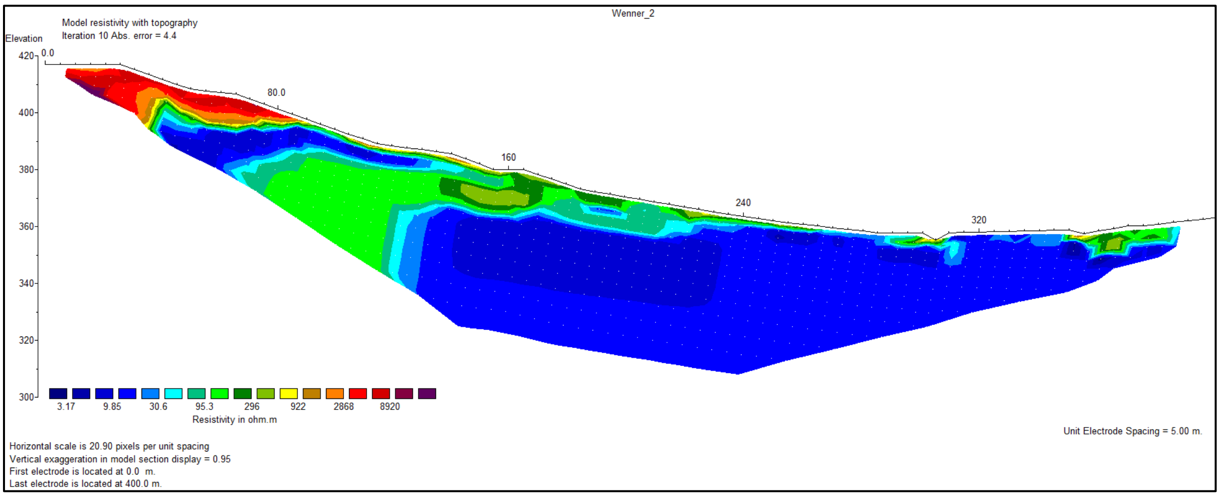

Figure 6: Ert profile with topography in the Heinersreuther Forst. The location of the profile is marked in the geomorphic map in Figure 4 (Own Figure; Source: Own measurements)

Figure 6 shows the robust ERT inversion results for the profile of the Heinersreuther Forst with the assessed topography. There are 4 layers visible in Figures 6 and 7. At an elevation of about 360 m the lowest layer ends. The values of this layer are lower than 50 Ωm. From profile meter 70 to 240 the layer is deformed and mixed with the above situated layer. The prevalent color here is green with values from 100 Ωm to 500 Ωm and it is 15 m thick. Above, a layer of varying thickness (15 m to 5 m) is situated and again deformed and interlocked with the layer below. Its values are like those of the lowest layer and exists from profile meter 30 to 140. On top, from profile meter 0 to 90, a layer of higher resistivity than 3000 Ωm with a varying thickness between 7.5 m to 20 m is located. Continuing from this part on is a 1 m to 2 m thick film of high resistivities to profile meter 275. The lowest points of the profile feature higher resistivity values (above 1000 Ωm).

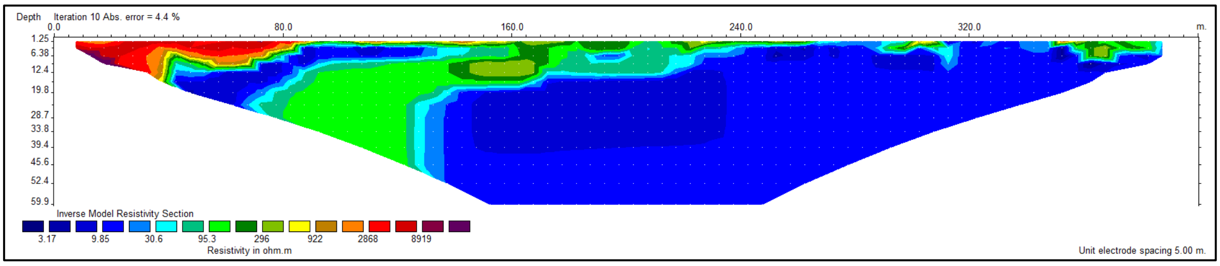

Figure 7 Ert profile in the Heinersreuther Forst. The location of the profile is marked in the geomorphic map in Figure 4 (Own Figure; Source: Own measurements)

Inversions of the maximum and minimum resistivity sections in Figure 8 are alike and just differ minimally from the inversion model in Figure 7. Therefore, the quality of our model can be assumed to be excellent. A RMSE of 4.4 % is a superb value, which suggest the same conclusion (Loke 2022).

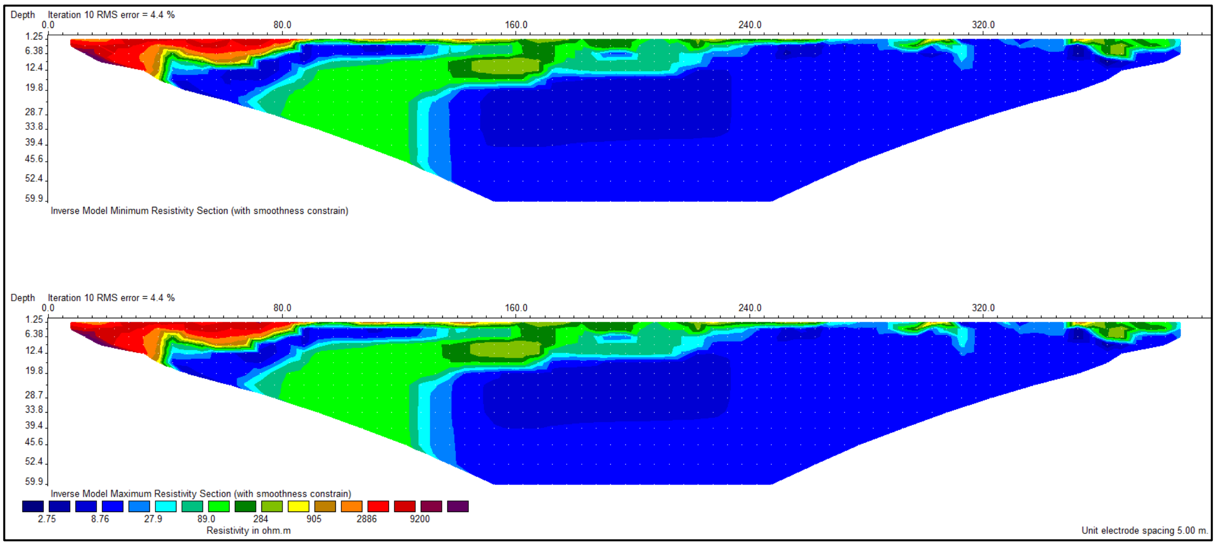

Figure 8: Minimum (top) and maximum (bottom) resistivity inversion results for the ERT profile in the Heinersreuther Forst. The location of the profile is marked on the geomorphic map in Figure 4 (Own figure; Source: Own measurements)

Uncertainties of the models are shown in Figure 9. Errors of up to 20 % occurred between layers. While 20 % are elevated levels of uncertainty, they are rarely observable in our model and the model can be considered as good (Loke 2022).

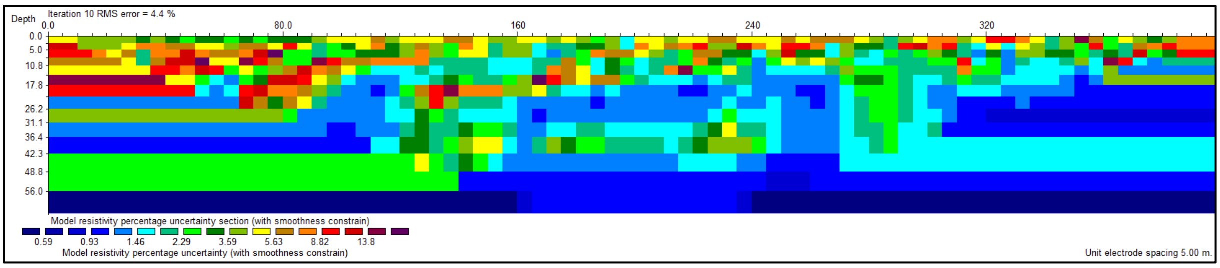

Figure 9: Uncertainties of the inversion model for the ERT profile in the Heinersreuther Forst. The location of the profile is marked on the geomorphic map in Figure 4 (Own figure; Source: Own measurements)

Based upon resistivity values and layering two slide types can be differed: A rotational slide from profile meter 40 to 100 and a subsequent translational slide to profile meter 270. Figure 10 presents the boundaries as well as slices of different failure plane angles.

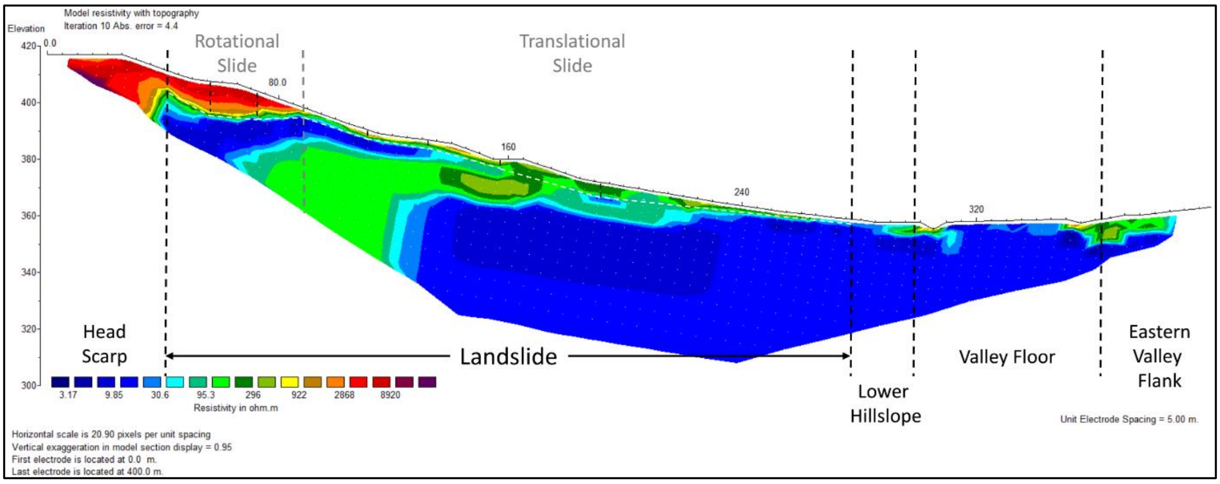

Figure 10: ERT with topography of the Heinersreuther Forst. A differentiation was made between head scarp, landslide, lower hillslope, valley floor, and eastern valley flank. The landslide part is further distinguished into rotational and translational slide and their slices (Own figure; Source: Own measurements)

The geology along the transect starts in the Rhätolias on top of the head scrap with an organic top layer of up to 5 cm (Figure 11f). This sandy soil with an organic overlay of varying thickness continues until the stream (Figures 11g-k). It is important to note that the moisture of the soil continuously increased towards the first stream, which was documented in Figures

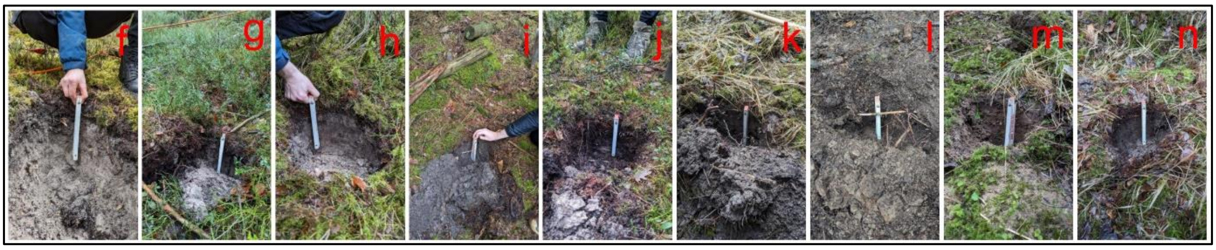

Figure 11: Soil tests along the ERT transect. From left to right probes of the profile meters 0, 50, 100, 150, 200, 250, 300, 350, and 400 are shown (Own Figure; Source: Own sampling)

11i and 11k with a dark gray excavated sand. Along the two streams a clayey sand or sandy clay is situated with no visible organic top layer (Figure 11l; Figure 11m). On the other valley flank a soil profile like in Figure 11k can be found in Figure 11n.

4.3 Results of the Root Analysis

The left diagram in Figure 12 presents the RAR values of the investigated species per depth class. European beech has the highest RAR in the uppermost and lowest section of the soil. Its value in a depth of 0 cm to 10 cm with 0.27 % is the highest overall. In the layers in between European beech is in between Norway Spruce and Scots Pine. Norway spruce has for all depths the lowest value for the RAR of all investigated species. The only exception is the top layer, where a RAR of 0.20 % is in between the other species. Finally, scots pine, which has the highest RAR values in depths of 10 cm to 40 cm. It is also the only species to reach its peak of the RAR in the middle layer. A RAR of 0.21 % is closely followed by the RAR in a depth of 10 cm to 20 cm with 0.20 %. Scots pine has the lowest value in the first section and is in between the other species in the layer at the bottom. The general trend throughout all species is observable that the RAR is higher in upper layers and sinks with increasing depth.

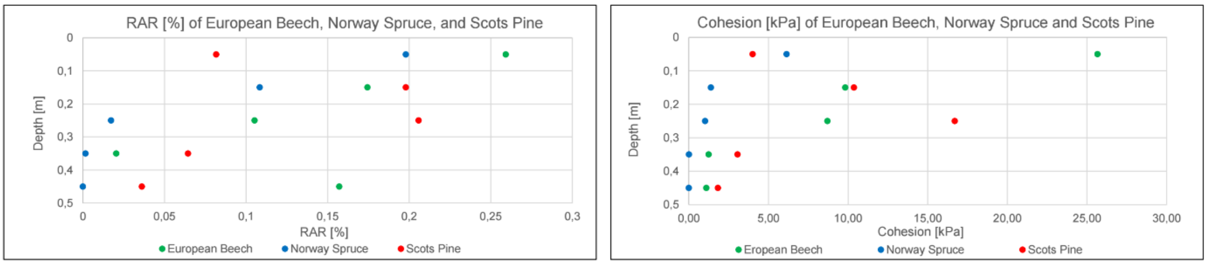

Figure 12: RAR in % and cohesion in kPa for european beech, norway spruce, and scots pine sampled from three representative trees (Own Figure; Source: Own sampling)

Figure 13 shows the T_rs of the investigated tree species. The highest T_r values are reached by european beech with up to 52 MPa. Smaller root diameters cause higher T_rs. For norway spruce the same trend is visible. While the T_r has the same trend, it is significantly lower than for european beech. The maximum for a breaking point diameter of 1 mm is 29 MPa. The scots pine has a contradicting trend. In comparison to logarithmic trends for european beech and norway spruce the scots pine has an anti-logarithmic function. In contrast to other species scots pine has its peak at a breaking point diameter of 2 mm with 23 MPa.

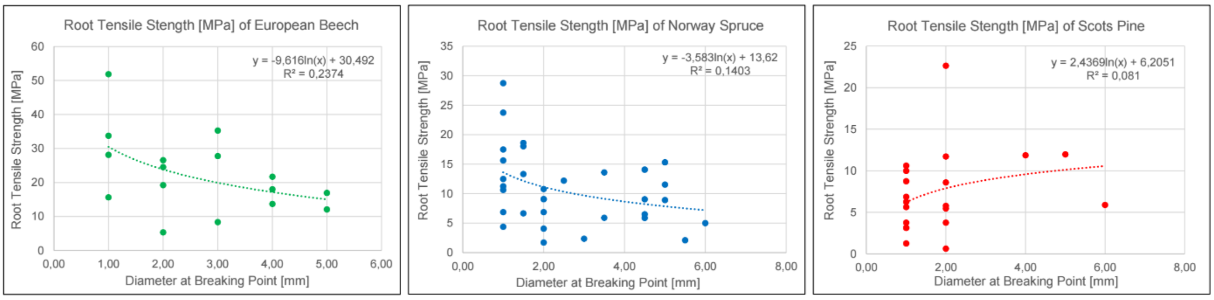

Figure 13: T_r in MPa for european beech, norway spruce, and scots pine plotted against the breaking point diameter (Own Figure; Source: Own sampling)

In the right diagram of Figure 12 the cohesion per depth class of the investigated tree species is shown. Observable trends are like the ones for the RAR. European beech is again strongest in upper layers with a cohesion of 25.5 kPa. The cohesion then decreases in lower layers. As for the RAR norway spruce has no impact in the lowest section.

For this study, an exceptionally small sampling size was used. While other studies like Bischetti et al. (2009) use statistically relevant sampling sizes, we used around 50 roots for the study on one investigated tree per species. Still, we were able to determine the same trends as previous studies but with halved values (Bischetti et al. 2009). Further problems of the method will be discussed in the subsequent chapter.

5. Discussion

While we were able to observe several features of a landslide scattered in the research area, just fallen or bent trees could have been recent. This is a first indication of a decreased landslide activity (Bierman and Montgomery 2014; Dräbing et al. 2023). However, it is possible to determine the FoS more accurately with the help of the ERT. There a layering becomes clear. The second and fourth layer must be claystones from the Feuerletten, with values below 100 Ωm (Dräbing et al. 2023). The third layer in between those is an interfering sandstone layer in the Feuerletten (ibid.). The uppermost layer is then the Rhätolias with the highest resistivity values in the ERT (ibid.). High resistivity values where the streams are situated result from eroded and deposited Rhätolias sandstones (ibid.). The landslide area takes up an area of 93,662 m^2. In ArcGIS we identified the length of the head scarp to be 396 m and the average length of the landslide to be 230 m (Figure 6). The rotational slide is 50 m long, and the translational slide is 180 m long. The average depth of the rotational slide is 9 m while the average depth of the translational slide is 3 m. Therefore, it is possible to calculate the volume of the rotational slide (178,200 m^3) and the volume of the translational slide (213,840 m^3), to get to the overall landslide volume of 392,040 m^3. For the stability analysis of the landslide the factor of safety was calculated for different scenarios of partial saturation (0.3, 0.6, 0.9) and full saturation (Bierman and Montgomery 2014; Dräbing et al. 2023; Selby 1993). The results for the distinct landslide parts are presented in Table 1.

Table 1: FoS values for the saturation levels of 0.3, 0.6, 0.9, and 1 for the rotational slide and the translational slide in the Heinersreuther Forst (Own figure; Source: Own measurements)

Saturation	FoS Rotational Slide	FoS Translational Slide
0.3	2.40	0.79
0.6	2.25	0.70
0.9	2.10	0.62
1	2.05	0.59

FoS values of the rotational slide are constantly above 1 and it strongly suggests a reduced activity (Bierman and Montgomery 2014; Dräbing et al. 2023; Selby 1993). Simultaneously, the exact opposite is visible for the translational slide. Its FoS is for all saturation levels below 1, which hints at an ongoing translational slide (ibid.). While we considered cohesion in the calculations it is normally increased in a forest by root cohesion (Bierman and Montgomery 2014; Dikau et al. 2019). Depending on species compositions and utilization forests can add up to 20 kPa of cohesion to the soil. Contradicting to our results, where beech caused the most additional cohesion, Bierman and Montgomery (2014) describe conifers as the tree inducing the highest cohesion. Roots can have an impact on cohesion a decade after clear cuts, the cohesion then sinks continuously to 0 kPa (Bierman and Montgomery 2014). Saplings of replanted forests post clear cut do not have the roots to replace adult trees cohesionwise, therefore after clear cuts a time window of 50 year opens an increased probability for landslides (ibid.).

Data Quality

Due to a tight time schedule the quality of the geomorphic map is lacking. Some features were mapped in the lab through remote sensing, which results in a less detailed and assured map (Seijmonsbergen 2013). ERT results are extraordinarily good. A RMSE of 4.4 %, the maximum and minimum models that look alike, except for minor details, and an uncertainty model which has maximum errors of 20 % with an average of 4 % are excellent quality measurements of the ERT (Loke 2022). It is important to remember that the ERT just registers pseudo sections of the sub surface. These are simplifications of the actual lithology and therefore cause uncertainties (Lange and Jacobs 2005).

The method for the root analysis is problematically inaccurate (Hales et al. 2013). Since we are not vegetation experts, other roots may have been sampled as well. The saturation phase had a negative effect on the testing. The bark got extremely slippery and caused the bark to slip off the wood, which led to an invalid try. A tendency for a rupture near the clamps was observed as well. This could be fixed by corking the ends of the clamps, which distributes the applied pressure in a better way (ebd.). The stability of the measurement structure and the weighing scale cause inaccuracies too (ebd.). This results in our study to extremely low R^2 values, which are even lower than in other studies probably because of sampling sizes (Bischetti et al. 2009; Hales et al. 2013).

6. Conclusion and Outlook

We were able to find the extent of the landslide area as well as its composition and activity. The landslide inflicted area comes up to an area of 93,662 m^2. Total material involved in the landslide was estimated to 392,040 m^3, with 178,200 m^3 allotting to the rotational slide and 213,840 m^3 allotting to the translational slide. Through the analysis of the FoS of the slide types it was possible to find evidence for a reactivated translational landslide while the rotational slide stays inactive. This is also what landslide features suggest. The quality of the study can be improved. A more detailed geomorphic map would be an interesting approach to deepen the knowledge about the area. The ERT result is already particularly good, but additional profiles rectangular to the profile can help to further investigate the landslide. The root analysis can be improved through an improvement in the method itself. As an expansion of the methods and research drill holes can give a better insight into geology. Also, pins can be installed into the soil and then precisely located per GPS or by their tilt measure the landslide movement especially in the translational part of the landslide.

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